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Today's battery bids can reproduce a state-of-charge market

Equivalent interval bids for energy storage, built from technical data and prices only

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Why it matters

A battery's value is intertemporal: what it sells now depends on what it bought before. **Baldick's SOC-based design** lets the ISO optimise the state of charge from the batteries' technical data, and is efficient. But it needs a market reform, moves battery management from owners to the ISO and is computationally heavier. Markets still clear **interval-by-interval bids**.

Research questions

RQ1 Can interval bids reproduce the SOC-based outcome, and what information do they need? *Answered here.*

RQ2 How much welfare is lost when the bids rely on imperfect price forecasts?

RQ3 Can an ISO build the same bids from its own forecast to monitor storage market power?

TAKE-HOME

Interval bids reproduce Baldick's SOC-based outcome exactly.

They need only the battery's technical data and its local prices. The gap between the designs is an **information gap**: price forecasts.

From Baldick's SOC market to today's interval bids

Current design · Baldick's SOC design · Our results · Use with forecasts

1 · MARKET CLEARING

Current design
ISO market-clearing LP · ERCOT-type day-ahead market
- Battery = paired *stepped* bid $(u_t, \hat{p}_t)$ to charge and offer $(c_t, \hat{g}_t)$ to discharge, **interval by interval**
- No SOC in the market: the owner manages it

Lagrangian relaxation · Prop. 2
dualise the SOC rows at the optimal $(\pi^\star, \bar{\nu}^\star, \underline{\nu}^\star)$
⇒ the problem separates by interval

same prices $\lambda^\star$ · same dispatch

Baldick's SOC-based design
ISO market-clearing LP with the batteries' state of charge
- Battery submits technical data $\theta = (\alpha, \beta, \gamma, \varepsilon, \zeta, \underline{s}, \bar{s}, \bar{p}, \bar{g}, s_0, w)$
- ISO optimises the SOC: dynamics $[\pi]$, SOC limits $[\bar{\nu}, \underline{\nu}]$, power limits $[\rho, \sigma]$
- Efficient benchmark, but needs a market reform

Equilibrium
KKT of an LP

Equilibrium
KKT of an LP

2 · EQUILIBRIUM

Current design · equilibrium
ISO clearing + each battery reacts per interval
- charge if $u_t \geq \lambda_t$, discharge if $c_t \leq \lambda_t$
- KKT: $\lambda_t - u_t = \underline{\rho}_t - \bar{\rho}_t$, $c_t - \lambda_t = \underline{\sigma}_t - \bar{\sigma}_t$
- Prop. 3: Baldick's schedule must lie on each battery's bid and offer curves at $\lambda^\star$

Equilibrium bid functions (E) · Prop. 1
$u_t = \alpha(\pi^\star_t - \bar{\nu}^\star_t) - \varepsilon$ $c_t = \zeta + \beta(\pi^\star_t + \underline{\nu}^\star_t)$
blocks $\hat{p} = \bar{p}$, $\hat{g} = \bar{g}$ ⇒ same KKT system, same $\lambda^\star$

Catch: (E) uses Baldick's duals, which only exist inside the SOC-based market.

Baldick's design · equilibrium
ISO clearing + each battery's self-schedule given λ
(S5) $\varepsilon + \lambda_t - \alpha(\pi_t - \bar{\nu}_t) + \bar{\rho}_t - \underline{\rho}_t = 0$
(S6) $\zeta - \lambda_t + \beta(\pi_t + \underline{\nu}_t) + \bar{\sigma}_t - \underline{\sigma}_t = 0$
(S7) $\pi_t = \gamma(\pi_{t+1} - \bar{\nu}_{t+1} + \underline{\nu}_{t+1})$, $\pi_T = w$
π = value of stored energy · $\bar{\nu}, \underline{\nu}$ = SOC scarcity rents

Bids from technical data + prices only
same outcome as Baldick (RQ1)

take the battery's part of the KKT

3 · OUR RESULTS

④ All equivalent bids
Prop. 5 · Cor. 5
The set $\Phi(\theta, \lambda)$ is a polyhedron: one self-schedule LP + linear conditions.
π is unique where prices pin it; at the kinks (battery full or empty) it is an interval, so we add a **selection rule**.

Compute: solve the battery's own small LP at $\lambda^\star$, read *any* optimal dual $(\pi, \bar{\nu}, \underline{\nu})$ and plug it into (E)

main path: all that is needed to build the bids

① Decomposition
Lemma 4 · Theorem 1
The battery's market duals are the duals of **its own self-schedule LP** at $\lambda^\star$.
Needs only θ + its own nodal prices: not other bids, the network or ramps. *Any* optimal dual works.

Interpretation: closed-form bids *(optional)*
Not needed to compute the bids (the LP already gives π and ν), but they show **which** prices matter and **how** a forecast error moves the bids.

same bids, closed form

same duals

③ Remove π · Prop. 4 · Cor. 4
marginal interval τ of a segment:
$\nu_\tau = (\lambda_\tau - \zeta)/\beta$ or $(\lambda_\tau + \varepsilon)/\alpha$
$\pi_t = \gamma^{\tau - t}\nu_\tau$ inside the segment
the highest charging price sets ν

② Remove ν · Cor. 3
rent $r_t = \pi_{t-1}/\gamma - \pi_t$
$c_t = \zeta + \beta \max\{\pi_t, \pi_{t-1}/\gamma\}$
$u_t = \alpha \min\{\pi_t, \pi_{t-1}/\gamma\} - \varepsilon$

in practice: replace $\lambda^\star$ by a price forecast $\hat{\lambda}$

4 · USE

Inputs
technical data θ
+ price forecast $\hat{\lambda}$
the owner's or the ISO's

Self-schedule LP
one battery, small
gives an optimal dual (π, ν)

Bids
$(u, c) \in \Phi(\theta, \hat{\lambda})$
via (E) + selection rule

A · Battery owner bids in today's market
Baldick's outcome without the reform. Perfect forecast $\hat{\lambda} = \lambda^\star$ ⇒ exact. Forecast errors ⇒ welfare loss ΔSW against Baldick's benchmark (RQ2).

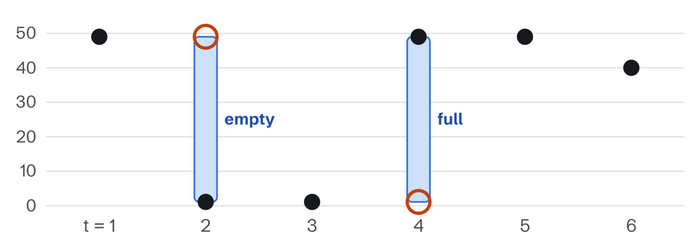
B · ISO monitoring (CAISO-type)
The ISO builds reference bids from its own forecast (as in RUC or a storage DEB) and compares them with the submitted bids to flag market power (RQ3).

Worked example: Baldick's base case

hour t	1	2	3	4	5	6
price $\lambda^\star$ ($/MWh)	43.1	50	0	0	50	50
charge $p^\star$ (MW)	100	0	200	22.2	0	0
discharge $g^\star$ (MW)	0	90	0	0	175	25
SOC $s^\star$ (MWh)	360	**0**	720	**800**	100	**0**
value $\pi^\star$ ($/MWh)	49	1.11	1.11	49	49	40
bid u_t ($/MWh)	**43.1**	**0**	**0**	**0**	**43.1**	**35**
offer c_t ($/MWh)	**50**	**50**	**2.11**	**50**	**50**	**50**

$\alpha = 0.9$, $\beta = \gamma = 1$, $\varepsilon = \zeta = 1$ $/MWh, $\bar{s} = 800$ MWh, $w = 40$ $/MWh, 4-h intervals. These bids clear today's market at the same prices and dispatch as Baldick's design.

The kink: π is not always unique



● solver dual ○ selection rule (pre-limit value) ▮ admissible range

When the battery is exactly empty ($t = 2$) or full ($t = 4$), any π in [1.11, 49] gives the same outcome. Prop. 5 gives the whole set; the rule picks one point instead of relying on the solver.

Next steps

RQ2 Build bids from price forecasts and measure the welfare gap ΔSW against Baldick's benchmark.

RQ3 Screen submitted bids against ISO reference bids $\Phi(\theta, \hat{\lambda}^{ISO})$, compared with a CAISO-type storage DEB.

References

1. R. Baldick, "A state-of-charge based formulation for storage participation in electricity markets: Technical reference," arXiv:2606.14993, 2026.
2. Y. Chen, R. Baldick, "Battery storage formulation and impact on day ahead security constrained unit commitment," IEEE Trans. Power Syst., 37(5), 2022.
3. CAISO Dept. of Market Monitoring, "Battery bid cost recovery and mitigation data," 2024.